Practice Set 4 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Discuss in brief, present concepts in various GU cancers

Type: LAQ (25 marks)

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).
- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive**

disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.

- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.
- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

Key points to remember:

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Q2. Robots in urology

Type: LAQ (25 marks) (also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP in **2000**; first paediatric robotic procedure **2002**.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) magnified view, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
S (2006) / Si	High definition, fourth arm, dual console for training
Xi (2014)	8-mm camera in any port, finer arms, docks from either side, no stirrups
SP (2018)	All instruments through one 2.5-cm cannula

- After patent expiry, other platforms appeared (e.g. **Hugo, Versius**, the Indian **SSI Mantra**).

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, less blood loss, shorter stay; equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable.
- **Retzius-sparing** and **hood** techniques preserve continence structures and give earlier continence.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy: shorter warm ischaemia and fewer conversions, so more complex and hilar tumours are done minimally invasively.

- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), less blood loss and transfusion.
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications.

4. Other procedures

Procedure	Why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children (small working space)
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery raises intracranial and intraocular pressure.

Complications specific to robotics

Problem	Details
Steep Trendelenburg	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position
Thermal bowel injury	Insulation failure, capacitive coupling; presents late
Robot malfunction	<1%; half found before the patient enters
Port-site hernia	Close larger port sites

- **Prevention**: pad pressure points, secure arms before tilting, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulators, animal and cadaver labs, then proctored cases. **FRS (Fundamentals of Robotic Surgery)** curriculum. Outcomes improve with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency).
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement.

> **Key points to remember:**
>
> - da Vinci 2000; Xi 2014; SP 2018; 7 degrees of freedom.
> - Yaxley: RARP equal to open at 12 weeks; RAZOR non-inferior; iROC more days out of hospital.
> - RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
> - Trendelenburg causes oedema and raised IOP; avoid shoulder braces.

Q3a. Newer PCNL techniques and nomenclature of miniaturised PCNL

Type: SAQ (10 marks) (mini-PCNL also answered in 2022-Summer Paper 1, Q1; prone versus supine in 2024-Winter Paper 1, Q3f)

Why miniaturise

- In PCNL (percutaneous nephrolithotomy), bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

Nomenclature (by tract or outer sheath size)

Name	Tract	Features
Standard	24–34 Fr	Ultrasonic or pneumatic lithotripsy; graspers
Mini	15–22 Fr	First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect)
Super-mini	About 10–14 Fr	Sheath with **active suction** of fragments
Ultra-mini	11–14 Fr	3 Fr inline telescope in 11–13 Fr sheath
Micro-perc	<10 Fr	"All-seeing needle" with optics; laser only, fragments pass spontaneously

- **Problem:** names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- **Working channel, not sheath size, decides what passes:** the smallest systems take only a laser fibre.

Newer techniques

1. **Supine PCNL and ECIRS (endoscopic combined intrarenal surgery):** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.
3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

Evidence

- Mini versus standard (meta-analysis): **non-inferior stone-free rate**, fewer transfusions, shorter stay; longer operation. Against a **24 Fr** tract these advantages disappear.

- Poor outflow through small sheaths raises **intrarenal pressure: fever, sepsis**.
- **Also know:** robotic or artificial intelligence-assisted puncture under study.

Diagram to draw: scale bar of tract sizes from micro to standard.

> **Key points to remember:**
> - Standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr.
> - Jackman 1998: first mini-perc (13 Fr).
> - Working channel decides instruments; suction lowers pressure.
> - Mini: fewer transfusions, longer time; equal to 24 Fr.

Q3b. VI-RADS

Type: SAQ (10 marks)

Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.
- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

Scoring

Score	Invasion	T2W	DCE	DWI
1	Highly unlikely	<1 cm, ± stalk; dark line intact	No early muscle enhancement	Muscle continuous
2	Unlikely	>1 cm with stalk and thick inner layer, or sessile; line intact	Inner layer enhances only	As 1
3	Equivocal	No category 2 features; line intact	No muscle interruption	As 1
4	Likely	**Line interrupted**	Enhancement extends into muscle	High signal into muscle
5	Highly likely	**Into perivesical fat**	Through wall into fat	High signal through wall and fat

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.
- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

Diagram to draw: bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

Key points to remember:

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Q3c. PI-RADS and MRI–ultrasound fusion biopsy

Type: SAQ (10 marks) (mpMRI and PI-RADS also answered in 2024-Winter Paper 1, Q3d; biopsy in Practice Set 4 Paper 2, Q3d)

PI-RADS

- **PI-RADS (Prostate Imaging Reporting and Data System)** v2.1 (2019) scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm^3 or EPE [extraprostatic extension]) on mpMRI (multiparametric magnetic resonance imaging): T2W (T2-weighted: anatomy), DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient: cellularity), DCE (dynamic contrast-enhanced: vascularity).

Score	Significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker at score 3
Peripheral	**DWI**	DCE positive → 4
Transition	**T2W**	DWI 5 → 4 (v2.1: T2W 2 with DWI ≥4 → 3)

- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **Also know:** PI-QUAL (image quality); PRECISE (active surveillance).

Fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS (transrectal ultrasound)** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

	Cognitive	Software fusion	In-bore MRI
Method	Aims from memory of MRI	Software overlay	Needle under MRI
Drawback	Operator-dependent; small lesion in large gland	Cost; registration error	Time, cost

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

Indications

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.

Evidence

- **PROMIS (2017):** MRI sensitivity 93% versus 48% for TRUS biopsy; 27% avoid biopsy.
- **PRECISION (2018):** MRI pathway found more significant cancer (38% versus 26%); 28% avoided biopsy.
- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

Diagram to draw: MRI target on live TRUS with needle path.

Key points to remember:

- Peripheral zone by DWI, transition zone by T2W; DCE only lifts peripheral 3 to 4.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.
- PROMIS and PRECISION: MRI before biopsy.

Q3d. Artificial intelligence in urology

Type: SAQ (10 marks)

Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

Development steps

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Data collection and labelling → training → internal validation
 → external validation → regulatory approval → use and monitoring
```

Applications

Area	Examples
Imaging	CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv)
Pathology	Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021)
Endoscopy	Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature
Stones	Predict composition and stone-free outcome
Prognosis	Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy
Robotic surgery	Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy
Paediatrics	Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models
Others	Urodynamics interpretation; LLM chatbots for patient information

Advantages

- Fast big-data processing; consistency; decision support.

Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

Diagram to draw: neural network: input layer (image pixels), hidden layers, output (stone or no stone).

Key points to remember:

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

Q3e. Indications of genomic testing in carcinoma prostate

Type: SAQ (10 marks) (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

Types of test

Type	Sample	What it finds	Use
Germline	Blood or saliva	Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2)	Therapy, family testing, screening
Somatic	Tumour tissue (usually primary) or ctDNA (circulating tumour DNA)	HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency)	Targeted therapy
Gene-expression classifiers	Biopsy or prostatectomy tissue	Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes)	Prognosis

Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).
- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
  - **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).
  - **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).

- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

Diagram to draw: flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

> **Key points to remember:**
>
> - Germline: all metastatic, high-risk, family history, Ashkenazi.
> - Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
> - BRCA2: aggressive; avoid casual active surveillance.
> - Decipher, Oncotype GPS, Prolaris: prognosis only.

Q3f. Gene therapy in urology

Type: SAQ (10 marks) (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

Vectors

Vector	Features
Adenovirus	Efficient, non-integrating, transient expression; replication-incompetent or oncolytic
Retrovirus, lentivirus	Integrate: long expression, risk of insertional mutagenesis
AAV (adeno-associated virus)	Low immunogenicity, long expression
Non-viral	Naked plasmid, liposomes, nanoparticles: safer, less efficient
Gene silencing	siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental

Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

Diagram to draw: adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

Key points to remember:

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Rapid Revision Sheet – Practice Set 4 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
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- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Q2. Robots in urology (LAQ)

- da Vinci 2000; Xi 2014; SP 2018; 7 degrees of freedom.
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- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
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Q3a. Newer PCNL techniques and nomenclature of miniaturised PCNL (SAQ)

- Standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr.
- Jackman 1998: first mini-perc (13 Fr).
- Working channel decides instruments; suction lowers pressure.
- Mini: fewer transfusions, longer time; equal to 24 Fr.

Q3b. VI-RADS (SAQ)

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Q3c. PI-RADS and MRI–ultrasound fusion biopsy (SAQ)

- Peripheral zone by DWI, transition zone by T2W; DCE only lifts peripheral 3 to 4.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.
- PROMIS and PRECISION: MRI before biopsy.

Q3d. Artificial intelligence in urology (SAQ)

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

Q3e. Indications of genomic testing in carcinoma prostate (SAQ)

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.

Q3f. Gene therapy in urology (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.

- Lumasiran: RNA interference for primary hyperoxaluria type 1.